## Multiplication of algebraic terms

**1** Simplify the following:

**a** $10 \times 6u$ **b** $7 \times (-3u)$ **c** $5x \times 2$ **d** $11 \times 3y$

**e** $(-8x) \times 9$ **f** $(-12) \times (-2u)$ **g** $(-3w) \times 2$ **h** $10 \times (-3y)$

**2** Simplify the following:

**a** $x \times x$ **b** $2x \times 3x$ **c** $4y \times y$ **d** $3a \times 5a$

**e** $8x \times 2xy$ **f** $9b \times 3b$ **g** $(10a)^2$ **h** $(4x)^2$

**3** Simplify the following:

**a** $9 \times m \times n \times 8$ **b** $w \times 4 \times z \times 6$

**c** $10 \times (-r) \times s \times (-5)$ **d** $9r \times 6s$

**e** $6u^2 \times 7v^8$ **f** $16p^3 \times 14q^3$

**g** $(-2a) \times (-4b)$ **h** $3w \times (-7z)$

**i** $(-5r) \times (-4s)$ **j** $(-10r^8) \times 6s^7$

**k** $4p^5 \times (-3q^5)$ **l** $2a^4 \times (-5b^3)$

**4** Simplify the following:

**a** $6r \times 2 \times 8s$ **b** $7w \times 9x \times 10y$

**c** $(-2w) \times (-4x) \times (-10y)$ **d** $(-8h) \times 5k \times (-3r) \times (-4s)$

**e** $5x \times 2y \times (-9)$ **f** $4a \times (-2a) \times 10a$

**g** $(-5x) \times (-2x) \times (-3y) \times (-3)$ **h** $(-2h) \times 4k \times (-j) \times (-5i)$

## Division of algebraic terms

**5** Simplify the following:

**a** $\frac{2x}{2}$ **b** $\frac{15v}{5}$ **c** $\frac{5m}{20}$ **d** $\frac{n}{4n}$

**e** $\frac{12xy}{12}$ **f** $\frac{63pq}{9p}$ **g** $\frac{12mn}{15m}$ **h** $\frac{6r}{rw}$

**i** $\frac{pr}{pqr}$ **j** $\frac{10u^6v^4}{u^6}$ **k** $\frac{-24a}{4}$ **l** $\frac{-11y}{y}$

**m** $\frac{y}{-11y}$ **n** $\frac{-12u}{3u}$ **o** $\frac{-4m}{-9m}$ **p** $\frac{-abc}{b}$

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q $\dfrac{k}{-jk}$ r $\dfrac{-6j}{jk}$ s $\dfrac{-ac}{abc}$ t $\dfrac{-2b^3}{3b^3}$

u $\dfrac{-3r^3w^5}{r^3}$ v $\dfrac{12n}{n}$

**6** Simplify the following:

a $5m \div 40$ b $20wz \div 4wz$

c $10r^6 \div 5r^6$ d $\left(-24r^4\right) \div 6r^4$

e $(-44rs) \div 4r$ f $(-36uv) \div (-6uv)$

g $10mn \div 5m$ h $18xy \div 6y$

i $27r^2 \div 9r$ j $\left(-20x^4\right) \div 10x^4$

k $\left(-22x^2y\right) \div -2xy$ l $(-50abc) \div (-5ab)$

**7** Simplify the following:

a $\dfrac{2x \times 3y}{xy}$ b $\dfrac{(-4x) \times 5y}{-10xy}$

c $\dfrac{6x \times 4xy}{8y}$ d $\dfrac{7x \times 4y}{2x \times y}$

e $\dfrac{10a \times 3b}{5b \times 2}$ f $\dfrac{12x \times 6y}{8x \times 2y}$

g $9x \times 4x \div 2x$ h $(-5x) \times 8y \div 10y$

i $11xy \times 3y \div y$ j $20y \div 4 \times 3y$

k $(10y \div 2y) \times (4x \div 2)$ l $(6x \times 3y) \div (9x \times 2)$

## Applications

**8** While Judy is packing rectangular boxes into crates, she notices that each crate is $12$ times wider than the width of one box, and $11$ times longer than the length of one box. Judy wants to know the greatest number of boxes she can pack into each crate.

Let the length of one box be $L$ cm, and the width of one box be $W$ cm.

a Find an expression for the volume of one box with a height of $44$ cm.

b Find an expression for the volume of a crate of height $H$ cm.

c Find an expression for the number of identical boxes that Judy can fit into each crate.

d If the crate is $88$ cm high, calculate how many boxes can Judy fit into each crate.

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**9** Aaron have books to be placed on a shelf which is $6$ times wider than the width of the book and $8$ times longer than the length of the book. Aaron wants to know the greatest number of books he can place into the shelf.

Let the length of one book be $x$ cm, and the width of one book be $y$ cm.

**a** Find an expression for the volume of one book with a height of $6$ cm.

**b** Find an expression for the volume of the shelf of thickness $z$ cm.

**c** Find an expression for the number of books that Aaron can fit into the shelf.

**d** If the shelf is $15$ cm thick, how many books can Aaron fit into the shelf?

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